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Design loops climb their scorer's error: an intervention-based audit for in silico molecular design

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Abstract

Closed-loop molecular design systems propose candidates, score them with a learned model, keep the best, retrain on the winners and repeat. Such loops are routinely reported by the improvement they produce on a held-out predictor. We show by direct intervention that this improvement does not measure molecule quality. Degrading the selection model on purpose, by withholding 80% of its training families while leaving the evaluation model untouched, made the reported gain **2.4 times larger**: +1.3182 against +0.5421, a difference of -0.7761 with a standard error of 0.0531 ($t = -14.60$, $n = 3$ seeds per rung, all six paired shuffled controls inside their pre-registered band). The breeder's equation $R = i \cdot \rho \cdot \sigma$, which is the usual justification for these loops, predicts the opposite sign. A deliberately worse scorer can only produce a larger apparent improvement if what the loop climbs is the scorer's error.

We then show why the standard defence, a second independent metric, usually cannot detect this. The gain reported by a k-winner mean carries an irreducible floor of $sd \cdot \sqrt{(2/k)}$. On our data (sd = 0.4562 per molecule, measured over 143 winner sets) that floor is 0.0645 at $k = 100$ against 0.0570 measured, and every value our own independent meter had produced, spanning +0.0138 to -0.0866, lies inside one such standard deviation of zero. Six reported configurations of "the independent metric did not move" were six measurements below the instrument's resolution.

We document four further measurement defects found in our own pipeline, each of which passed every check that existed at the time: a constraint that pinned a histogram while the continuous statistic beneath it drifted; a cross-validation splitter that was not the algorithm its own documentation described, leaving 20.1% of held-out molecules with a near-twin in training; a diversity cap on pairwise identity that cannot bound distributional narrowing; and a rank correlation that did not correct for ties. We close with a seven-step audit protocol and a register of every claim this programme has withdrawn.

We make no claim that our loop produces better molecules. It does not. On six seeded replicates the treatment arm sits 0.0652 below its own shuffled control on the independent meter (SE 0.0201, about 3.2σ). The contribution of this work is the instrument, not the gain.

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1. Introduction

The argument for closed-loop in silico design is an analogy to directed evolution. A generator proposes, a scorer selects, the generator is retrained on what was selected, and the population improves across generations. The quantitative form usually invoked is the breeder's equation,

$$R = i \cdot \rho \cdot \sigma$$

where R is the response per generation, i the selection intensity, ρ the correlation between the scorer and the truth, and σ the standard deviation available in the population. The equation predicts that response is **linear and increasing in scorer accuracy**: a better scorer should buy a bigger gain.

That prediction is testable without any new experimental data, and it is testable as an intervention rather than a correlation. One can degrade the scorer on purpose, hold everything else fixed, and watch what the loop does. To our knowledge this test is not standard practice. We ran it, and the sign came back inverted.

The remainder of this paper is organised around what that inversion implies, and around the question of why it had not been detected: the second metric used to detect exactly this failure is usually operating below its own resolution.

1.1 What this paper is and is not

This paper reports **computational measurements only**. No molecule designed in this programme has a measured potency or safety value. Every activity figure is the output of a ridge regression over descriptors or protein language model embeddings. Where we write "gain", we mean a change in the mean prediction of a held-out model, not an improvement anyone has observed in a laboratory.

The results are presented as a methods contribution for anyone running a score-and-select loop over molecules. They are not a drug discovery result.

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2. System and instruments

2.1 The loop

Each generation samples a candidate library from a small autoregressive peptide language model (0.81M parameters), scores every candidate, selects k winners under a composition constraint, writes them to disk, and retrains the generator on the winners by reward-weighted maximum likelihood with a KL anchor to the base policy. Default configuration: 20,000 candidates per generation, k = 100 or k = 200 winners, 8 or 12 generations, KL weight β = 0.5.

Every arm is run twice: a **real** arm with the true reward and a **shuffled** arm in which the reward vector is permuted within each generation. The shuffled arm is the paired control, and it is seeded identically so the two differ only in the ordering of the reward.

2.2 The scorers and the meters

Two model classes are used, and the distinction between them carries the paper:

role	model	fitted on
selection scorer	ridge over physicochemical descriptors, or ridge over ESM2 embeddings	half A of the labelled set
meter (descriptor class)	ridge over the same descriptor basis	half B, never touched by selection
meter (language model class)	ridge over ESM2-35M embeddings	half B
meter (instance based)	identity-weighted nearest labelled neighbours	half B, no feature space at all

Halves A and B are split by sequence identity family, so no family appears in both. The third meter exists because a held-out split of the *same* feature space is not independent in the sense that matters: a score can only be gamed in a basis it shares, and two ridges over one descriptor set share a basis.

2.3 Pre-registration

Every arm in this programme has a pre-registration file written before the run, naming its gates, its control tolerances and a prediction. Several of those predictions failed, and section 10 lists them. We regard the pre-registration files as the primary evidence that section 10 is complete rather than curated.

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3. Result 1: the gain is anti-correlated with scorer accuracy

3.1 Design

Two rungs, three seeds each, real and shuffled paired, twelve arms in one allocation so that every arm saw the same node and the same libraries.

The intervention withholds whole identity **families** from the two selection models, not random rows. Dropping rows inside a family leaves that region of sequence space covered, which is not a real loss of accuracy; dropping whole families removes coverage. The starved rung received 20% of the families, the full rung all of them.

Critically, **only the two selection fits were starved**. Both half-B meters were left intact, so both rungs are read on the same yardstick. The run script counts the fits it modifies and aborts if the count is not exactly two, because a run that silently starved a meter would place a false point on the axis and invalidate the comparison.

3.2 Result

rung	families given to the scorer	drift-corrected gain	sd	n
starved	20%	**+1.3182**	0.0848	3
full	100%	**+0.5421**	0.0358	3

Difference -0.7761, SE 0.0531, t = -14.60.

All six shuffled controls fell inside the pre-registered ±0.11 band, spanning -0.0851 to +0.0396, so the control conditions registered in advance hold.

3.3 Reading

The registered reading rule said that gain scaling with accuracy would mean labels are the lever and the machinery is sound, and that gain flat in accuracy would mean the machinery cannot compound. The measured outcome was a third branch we had not registered: the gain **rose** as the scorer got worse, at a magnitude far outside the registered power of 2.15σ.

A deliberately worse scorer has more error available to climb. A more accurate one has less. If the quantity being maximised were molecule quality, accuracy could not hurt. We therefore read the reported gain as a measure of **how exploitable the scoring model is**, and not as a measure of the molecules.

This also explains, in one mechanism, four observations we had previously treated as separate: the independent-class meter failing to follow the descriptor meter; internal diversity collapsing monotonically while the shuffled control holds flat; the held-out meter turning over mid-run; and the loop improving its own scorer's meter by +0.46 to +0.81 across four separately controlled arms.

3.4 What this result does not establish

The **mechanism** of the inversion is not established. A plausible account is that the selection scorer and the descriptor meter share a feature basis, so a starved scorer pushes the population further along a direction the intact meter also rewards. This run does not test that account. What is established is the sign and its size.

Nothing here shows the molecules are worse. It shows the meter cannot tell.

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4. Result 2: the resolution floor of a k-winner meter

4.1 Derivation

An independent meter reports the mean predicted value over the k molecules a loop kept, and the reported gain is a difference of two such means. If the molecules entering each mean are a sample, that difference carries

$$sd(gain) = sd_per_molecule \cdot \sqrt{(2 / k)}$$

by construction, independent of anything the loop did.

4.2 Measurement

Measured across 143 saved winner sets, sd_per_molecule = **0.4562**. At k = 100 the predicted floor is **0.0645** against **0.0570** measured over three fresh seeds, agreement within what an sd from n = 3 can resolve.

k	floor on the gain	seeds to resolve a 0.03 effect at 2σ
100	0.0645	about 18
200	0.0456	about 9
1000	0.0204	about 2
3250	0.0113	1

4.3 Consequence

Every value our independent meter had produced across six configurations, namely +0.0138, -0.0182, -0.0244, -0.0373, -0.0482 and -0.0866, lies inside one such standard deviation of zero.

We had been describing that column as "the loop fails to transfer to an independent class". The correct description is **"this meter at this k cannot tell"**. The distinction matters: the first is a finding, the second is an unresolved measurement, and only the second is true.

We note that this failure mode is not specific to our system. Any design loop that reports a gain on a held-out model evaluated over its top-k selections, with k in the hundreds and an effect size in the hundredths, is in the same position, and the floor is computable from quantities the authors already have.

4.4 The fix is design, not more runs

Raising k reduces the floor as √(1/k). Better, **pair the comparison**: score the same candidate library under both the real and the scrambled reward, so that the two arms differ only in which candidates they selected and the library-level sampling noise is common to both and cancels. The real arm already benefits from an implicit pairing against its own generation 0, which is why it reproduces far more tightly than its control (section 5).

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5. Result 3: treatment and control have structurally different variances

Over three fresh seeds at β = 0.5, 12 generations:

arm	seed 1	seed 2	seed 3	mean	sd
real	-0.0411	-0.0269	-0.0343	**-0.0341**	**0.0071**
shuffled control	+0.0962	-0.0174	+0.0422	**+0.0403**	**0.0570**

The control is eight times noisier than the treatment on the same meter.

The mechanism follows from section 4. Best-first selection is near-deterministic given the score landscape, and the landscape does not change with the seed, so the real arm's winners land in the same region every time. The shuffled control re-draws a fresh random k each time and pays the full sampling variance of a k-sized mean.

The practical consequence is a rule we had been violating: **neither arm's spread is the other's error bar.** Using the control's spread as the noise bar for the treatment, which is the intuitive move and the one we made, is wrong here by an order of magnitude.

Pooling all six seeded replicates across two anchor settings:

	mean	sd	n	SE
real arm	**-0.0331**	0.0219	6	0.0089
shuffled control	**+0.0321**	0.0440	6	0.0180
difference	**-0.0652**			**0.0201, about 3.2σ**

So the real arm sits *below* its control on the independent meter, supported at about 3.2σ on six seeded replicates. We flag that this is an arm-versus-arm comparison on one ruler, which is the comparison type that survives the fold defect in section 7; the absolute levels do not.

Note also what produced this result: not a new experiment, but replicates on a number that already existed.

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6. Result 4: a binned constraint cannot pin a continuous statistic

Prior work in this programme established that composition drift is the dominant failure of peptide optimisation: four separate routes all improved a score by moving net charge. We therefore pinned the net-charge distribution of the winners to the reference distribution by apportionment.

The pin passed its own diagnostic at every generation while the quantity it existed to protect drifted by 0.35 of a charge unit.

generation	0	1	2	3	4	reference
real arm median charge	3.15	3.00	**2.65**	2.75	2.85	3.00
shuffled arm median charge	3.00	3.00	3.00	3.00	3.00	3.00

The cause: net charge is **continuous**, because histidine carries a partial charge at pH 7.4. Across the 39,448 reference sequences, 30.1% carry a non-integer charge. The strata snapped charge to the nearest integer, so the stratum labelled "3" spans roughly 2.5 to 3.5, and best-first selection walks to the low edge of every stratum while the integer histogram stays exactly on target.

Three properties of this defect generalise:

1. **No histogram diagnostic can catch it.** A `pin_histogram_exact` check passes a run leaking 0.4 of charge. This survived a dry pass, a real run and a written result.
2. **The control looks healthier than the treatment.** The shuffled arm held 3.00 exactly, because it has no selection pressure to walk with. A control that outperforms the real arm on a constraint diagnostic is this bug's signature, not evidence of a healthy control.
3. **The summary statistic hid it too.** The gate computed median(last) − median(first), so a run sitting 0.3 off the reference at *every* generation reports a drift of 0.0.

The fix is two nested largest-remainder apportionments: the integer level stays exact so strata do not fragment at small k, then a second apportionment inside each stratum at a sub-bin width of 0.2. Winners inside one fine bin differ by at most 0.2, so the realised median cannot sit further than 0.1 from the reference's: the tolerance is met **by construction** rather than by moving the bar. With the nested pin, the median delta against the reference is 0.0000 in every generation of every arm reported here.

The generalisation: a constraint expressed at one resolution cannot bound a statistic defined at a finer one, and the diagnostic built from the constraint will not reveal the gap. We have now found four instances of this single error shape in one codebase.

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7. Result 5: audit the split, not the splitter

The function that builds our cross-validation folds documents itself as single-linkage clustering at an identity threshold. It is not. It performs greedy representative assignment where the first match wins, at a threshold of 0.6 against the 0.80 rule the evaluation is meant to mimic.

Measured directly, on the quantity the guard exists to bound:

set	mean cross-fold max identity	p95	max	fraction above 0.80
potency labels	0.671	0.923	0.986	**20.1%**
safety labels	0.658	0.933	0.971	**24.6%**

Families spanning more than one fold: 0 of 826 and 0 of 336. The test that existed passed perfectly.

Rebuilt with true single-linkage at 0.80, the cross-fold fraction above 0.80 is 0.0000 by construction, and accuracy falls:

model	potency, honest	potency, leaky	safety, honest	safety, leaky
descriptor ridge	0.4514	0.4596	0.3254	0.3901
ESM2-35M frozen	0.4646	0.4735	0.4026	0.4624
ESM2-650M frozen	0.4640	0.4856	0.3860	0.4081

The inflation reaches **+0.065** on the safety axis.

Both columns of that table are read on the same ruler, so the inflation is the quantity it measures. The absolute values in it are not quotable, for a reason unrelated to the leak: they were produced by the pre-correction harness, which chose its ridge penalty by inner cross-validation and scored with the uncorrected rank statistic of section 9. Rebuilt on the fixed ruler — one set of single-linkage folds written to disk once, the penalty averaged over rather than selected, and the tie-corrected statistic — the honest absolutes are:

model, honest folds and corrected ruler	potency	safety
descriptor ridge, 36 features	0.4502	0.4074
ESM2-35M frozen, mean-pooled	**0.4588**	—
ESM2-650M frozen, mean-pooled	0.4164	0.3502

The safety figure moves most, from 0.3254 to 0.4074, and almost none of that is the tie correction: the inner-cross-validated penalty swung four decades across five folds and one bad fold had been dragging the pooled score down. Deleting the selection step was worth more than any change of representation we tried.

The withdrawn fine-tuning result survives the honest re-run, and we report that it does. It had been measured at 0.5219 on the leaking folds and withdrawn on the reasoning that a 650M- parameter model has the capacity to memorise a near-twin's label where a 36-dimensional ridge does not. On folds with zero cross-fold pairs above 0.80 the same arm reads **0.5356 on potency and 0.5079 on safety**, slightly *higher* than the leaky reading, with a shuffled-label null of **−0.0228** at identical settings. So the fold leak was not what produced that gain. The withdrawal was the correct call on the evidence available at the time and the result itself holds; we record both, because a register that only ever loses claims is as uninformative as one that never does.

The structural lesson is the same shape as section 6. The clustering had a docstring that stated the right guarantee, a threshold argument that looked deliberate, a name that described the guarantee, and a test that families never span a fold, which passed. **Every check that was run passed, because none of them measured the quantity the guard exists to bound.**

7.1 The same defect in three published benchmarks

Everything above is a defect in our own code, which invites the obvious objection: this is a bug in one pipeline, not a statement about the field. We therefore ran the same measurement against third-party published splits that we had no hand in producing. Every file below was downloaded from its own published location and hashed at run time; the instruments are in the repository of section 14.

The measure is the one used throughout this paper: a Levenshtein ratio of at least 0.80 between a held-out sequence and any training sequence.

published split	homology control as published	exact overlap	twin >= 0.80
LRGB Peptides-func (NeurIPS 2022)	none; stratified random	7.72%	**65.29%**
AMP Scanner v2 positives (2018)	CD-HIT, 90% identity	0.00%	**16.71%**
AMP Scanner v2 negatives (same paper)	CD-HIT, 40% identity	0.00%	**0.00%**
PepBenchmark `hybrid_split` (ICLR 2026)	k-mer then MMseqs2, 30% identity	0.00%	**29.97% / 48.56%**

For LRGB both the data and split MD5s match the values hard-coded in the benchmark's own loader, so that row is exactly reproducible rather than approximately. The PepBenchmark figures are the mean over its five shipped seeds on `antimicrobial_E_coli_mic` and `hemolytic_hc50` respectively.

The AMP Scanner v2 pair is an instrument validation and should be read first. It is one paper, one tool and one pipeline, applying CD-HIT at 90% to its positive class and at 40% to its negative class before splitting. Our measure returns a sixth of the test set on the first and exactly zero on the second. A check that fired on both would be a false-positive generator; this one discriminates on threshold alone. It also localises the damage, because the contamination sits entirely on the positive class, and leakage confined to positives inflates sensitivity specifically rather than accuracy uniformly.

PepBenchmark is the load-bearing row. LRGB applies no homology control, so a finding there says only that it was not attempted. PepBenchmark deduplicates during cleaning and ships a 30%-identity split as its documented default, so it is a case where the control was applied and still did not arrive: going from a random split to the default removes between half and two thirds of the contamination and leaves 29.97% and 48.56% behind. Exact overlap is 0.00% on every dataset, strategy and seed, so the deduplication works exactly as described and our duplicate check correctly finds nothing. The residue is near-duplication, and its cause is the same one as in our own splitter: representative-based clustering compares each sequence to cluster representatives rather than to every member, so transitive chains of similarity are under-merged.

Three constraints on how far this can be pushed, each of which a reader will test:

4. **A contamination rate is not an inflation.** We retrained nothing on corrected versions of these splits, so what the leakage is worth in reported accuracy on their models is unmeasured. The only inflation figure in this paper is ours, and it does not transfer.
5. **For LRGB, only 30 of the 180 exact sequence matches share a SMILES string.** For a sequence model all 180 are leakage; for the graph models LRGB is used to benchmark, the remaining 150 differ in stereochemistry or modification and are near-duplicates. We state it as a shared primary sequence and never as a duplicate molecule.
6. **Redundancy reduction and homology-aware splitting are different operations.** AMP Scanner's 40% filter removed redundant sequences from the pool before splitting, so nothing similar survived anywhere; PepBenchmark assigns clusters to splits. The first can reach zero and the second cannot do better than its clustering. The two results are not in tension.

The defect in section 7 was therefore not ours alone, and it is not confined to groups who did not think about it. It survives in a benchmark whose authors identified the problem, chose a tool intended to solve it, and documented the threshold they used.

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8. Result 6: a pairwise cap cannot bound distributional narrowing

Internal diversity collapses monotonically in every real arm while the shuffled control holds flat, by -0.22 to -0.30 depending on the measure, with the control moving by less than 0.01.

We first addressed this with a maximum pairwise identity cap at 0.80, fully enforced: 112 internal rejections, 0 waivers over 900 winners. It removed about **12%** of the narrowing. The winners finish at mean pairwise identity 0.55 **with no single pair above 0.80**, which satisfies the cap completely.

Replacing it with a bound on the *mean* pairwise identity at 0.35 removes about **82%** of the collapse, with cap waivers at 0.9% of winners and the charge pin still exact.

constraint	identity rise over the run
none	+0.2945
max pairwise identity ≤ 0.80	+0.2606
mean pairwise identity ≤ 0.35	+0.0522

Our pre-registered prediction said the cap would pass the diversity gate, because we believed the collapse was caused by selecting near-copies. It is not. A tight cluster entirely below the pairwise threshold is a diversity collapse that no pairwise cap can see.

And with the collapse fixed, the independent meter still did not move. The diversity-bounded arm reports -0.0482 on the independent meter against a +0.10 bar. So diversity collapse was not the mechanism either.

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9. Result 7: a rank statistic that is not a function of its data

Every accuracy figure in this programme was a Spearman correlation, and for most of the programme's life it was computed by

```
ra = np.argsort(np.argsort(a))          # ordinal rank, ties broken by sort order
rb = np.argsort(np.argsort(b))
pearson(ra, rb)
```

Spearman's rho requires midranks: tied values share the mean of the ranks they span. Ordinal ranks instead give each tied value a distinct consecutive integer, in whatever order an unstable sort emits it. The statistic is therefore biased, because the tied block is given an internal ordering that correlates with nothing — and, worse, it is **not a function of the data**, because permuting the rows of the same table changes the answer.

Both label sets tie, and not by rounding. They tie because assays are read on doubling dilution series and because the safety assay is censored at its top dose.

target	n	distinct values	rows inside a tied	largest single block

			block	
potency, log10 MIC	2904	2349	26.1%	40 (1.4%)
safety, log10 HC50	1238	667	**51.2%**	**445 (35.9%), the censoring ceiling**

More than a third of the safety label is one censored value, "no haemolysis observed up to the top dose tested". The old statistic was assigning 445 arbitrary distinct ranks inside it.

Every model scored on the fixed ruler had already saved its pooled out-of-fold predictions, so the recomputation required no refitting: the same predictions were scored against the same labels twice, changing the rank convention and nothing else. **40 models.**

	potency, 23 models	**safety, 17 models**
corrected minus uncorrected	+0.0000 to +0.0000	+0.0015 to **+0.0090**
spread of the uncorrected statistic over row permutations of the same table	0.0002	up to **0.0387**
model pairs whose order changes under the correction	**0 of 253**	4 of 136

Three readings, in order of how much they matter.

The bias is small and it points the safe way. Every delta is zero or positive, so the uncorrected statistic was conservative. No absolute figure this programme published was inflated by its tie handling, and the largest correction anywhere is +0.009.

It reorders nothing. No potency pair changes order. The four safety pairs that do are separated by at most 0.0036 on the old ruler and 0.0023 on the new one, which is inside the statistic's own row-order spread: they were unresolved before the correction and they are unresolved after it.

And the finding we had not registered is larger than the one we had. On safety, re-sorting the rows of an identical table moves the old statistic by up to 0.039. Several comparisons we made on that axis are smaller than that number. They survive — both sides of each comparison were scored against a common label vector in a fixed row order, so the arbitrary within-tie ordering is shared and cancels, which is the same structure that protected the arm-versus-arm comparisons from the fold leak of section 7. What does not survive is quoting any pre-correction safety figure as a standalone accuracy at four decimals, which we did repeatedly.

The generalisable point is the one this paper claims for every defect in it. We had been auditing the models. The ruler was wrong, in a way that is one line of library documentation deep, and it had been wrong under every number every model produced.

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10. Register of withdrawn claims

We regard this section as load-bearing. A programme that reports only its surviving results gives the reader no way to estimate how many did not survive.

Withdrawn as measurement artefacts of the defects above:

- Our first headline, a held-out gain of +0.8357, withdrawn after its own control and then the independent meter overturned it.
- An entire run voided under our own kill rule for the charge drift in section 6, including its +0.6924 gain, recorded here only so that it cannot later be presented as evidence.
- A second arm voided for a median charge delta of -0.20, including its +0.6579 reading on the instance-based meter, which is therefore not evidence and is not read as any.
- A fine-tuning accuracy gain, withdrawn pending the honest folds of section 7. **Reinstated.** The honest re-run reads 0.5356 and 0.5079 against 0.5219 and 0.4678 on the leaking folds, with a shuffled-label null of −0.0228, so the leak was not what produced it. We leave the entry here rather than deleting it: a register that only records losses would misrepresent what withdrawal pending a measurement means.
- Absolute accuracy figures throughout this programme, pending recomputation with a tie-corrected rank statistic. Arm-versus-arm comparisons on a single meter are unaffected. **Resolved, in the direction that costs nothing** (section 9): the uncorrected statistic was conservative everywhere, by +0.000 on potency and at most +0.009 on safety, and it reorders no pair that was ever read as a difference. What is withdrawn, permanently, is the *precision* of every pre-correction safety figure: that statistic moves by up to 0.039 on row order alone, so those numbers never had the decimals we gave them.

Withdrawn as unreplicated, after seeded replicates:

- "A stronger KL anchor moves the improvement peak later and fixes the mid-run turnover." Replicated peaks were 5, 5, 6 against 8, 6, 6, overlapping sets. We had reported a lucky seed as a result.
- "$\beta = 2.0$ is the best setting." It is not; the weaker anchor gains more, reliably, with disjoint replicate ranges.
- A claimed replicate standard deviation of 0.082 to 0.105, which turned out to be configuration variation masquerading as seed noise, 5 to 10 times the true replicate spread.

Withdrawn as unmeasurable, the bar being below the instrument's noise:

- A gate on trajectory shape requiring the fall from peak to stay under 0.02, when a shuffled control with nothing to learn wanders 0.134 within a single run. Zero of six replicated arms passed it, including the controls.
- Two control tolerances of ±0.06 and ±0.03, both below the measured drift, withdrawn as mis-set rather than claimed as met.
- A decision gate whose two triggering thresholds each sat about 3.5 standard deviations from where the statistic actually lives. A gate that can only fire on a 4σ excursion is a formality, not a gate. One shuffled control scored within 0.004 of the threshold meant to declare the project alive.

Withdrawn as unverifiable:

- An early run's constraint verdict, because its medians were recorded to one decimal place and its winner sequences were overwritten by a later run. The sequences needed to recheck it no longer exist.

Refuted by direct measurement:

- The breeder's-equation framing for this loop (section 3).
- "The diversity collapse is variance exhaustion." Variance retains 90% of its generation-0 value while sequence diversity falls 29%, so the collapse is not a variance collapse.
- "Net charge explains the disagreement between model classes." Both classes under-price the true charge slope by about the same amount. Charge has now failed as an explanation by three independent methods.

- "The class-disagreement offset is a property of the small generator." Raw generator samples score *higher* on class agreement than real peptides do. The drop happens at the first selection step.

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11. An audit protocol

Each step below caught a result we had already written down.

7. **Run a starved-scorer rung.** Degrade the selection model deliberately, hold the evaluation model fixed, and verify programmatically that only the selection fits changed. If the reported gain does not fall, the gain is not measuring quality.
8. **Read the loop with a different model class**, and preferably with an instance-based method that has no feature space to share. A held-out split of the same basis is not independent.
9. **Compute the resolution floor before reporting a null.** $\mathrm{sd}\cdot\sqrt{(2/k)}$ is available from quantities you already have. A reading inside it is an unresolved measurement, not a negative result.
10. **Pair the comparison** on a shared candidate library so the sampling noise cancels, rather than adding seeds against an unpaired control.
11. **Price two noise floors separately**: within-run wander for any claim read off a trajectory's shape, and across-seed endpoint spread for any claim about a gain. They differ by an order of magnitude and a noisy path does not invalidate a measured displacement.
12. **Audit the split and the constraint, not the function that builds them.** Measure the quantity the guard exists to bound, directly, on the output. A passing test of a different quantity is the failure mode, not the assurance.
13. **Permute the rows and recompute.** A statistic worth quoting is a function of the data. Re-sort the table and score it again; if the number moves, the digits that moved were never yours. This costs one line, it is the cheapest step here, and it is the one that found the largest defect in our ruler.

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12. Limitations

- **n = 3 per rung** in the central experiment. The registered power was 2.15σ against the effect the breeder's equation predicted; the observed effect is far outside that, so the sign is not in doubt at n = 3, but the magnitude is loosely estimated.
- **One system.** Peptides, one generator family, two scorer classes. We do not show that the inversion generalises to other molecule classes, though the resolution floor in section 4 is arithmetic and does.
- **No experimental validation.** No molecule in this programme has a measured potency or safety value. Every number is a model prediction.
- **Absolute accuracy figures rest on one rebuilt ruler.** Both corrections have landed, but every absolute in sections 7 and 9 now comes from a single fold file and a single scoring path. The fold construction has been reproduced independently, by two implementations written on different machines that agree on 1198 and 450 clusters; the scoring path has not.
- **One number in section 7 we could not recompute ourselves.** The fine-tuned readings, 0.5356 and 0.5079, were produced on a GPU allocation and their out-of-fold predictions were not retained locally. They were scored by the corrected path, which we read from the code rather than re-measured.
- **The safety label is 36% censored** (section 9), so a rank correlation against it is partly a two-class problem. We report Spearman because the programme's earlier figures are Spearman and must remain comparable, not because it is the right statistic for a censored label.

- **The inversion mechanism is untested** (section 3.4).

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13. Methods

Labelled data are public antimicrobial peptide measurements, split into halves A and B by identity family. Selection scorers are fitted on half A, meters on half B. Spearman correlations are computed with whole families held out.

Candidate generation samples a 0.81M-parameter autoregressive peptide model. Selection applies a nested largest-remainder apportionment over net charge (integer level, then a 0.2-wide sub-bin level) with a novelty gate on maximum identity to the reference set. Retraining is reward-weighted maximum likelihood with a KL anchor to the base policy.

The starvation intervention replaces the fit method of the selection models with a wrapper that subsamples whole identity families, counts the fits it modifies, and raises an error if the count differs from the expected two.

All computation ran on a university HPC allocation at no cost. Job identifiers are recorded in the run logs for every figure quoted.

13.1 Disclosure of AI assistance

This research was conducted with substantial AI assistance. Large language models were used as the research engineering layer: writing and auditing analysis code, drafting pre-registration files, executing experiments on the compute allocation, and checking reported numbers against their source tables. Several of the corrections in section 10 were produced by one model auditing another's code, including the fold-clustering defect in section 7 and the tie-handling defect. The author directed the research, set the questions and gates, and is responsible for the content. This disclosure is made because the result in section 3 is about the reliability of automated selection, and the reader is entitled to know the same question applies to how this paper was produced.

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14. Availability

Code, pre-registration files and result files for every number quoted are archived at `https://github.com/braxtonmensah/silicyte-design-loops`, release tag `v1.0.0`. A Zenodo DOI for that release is pending and will be added to this sentence once minted. Each result file carries the job identifiers of the run that produced it.

The repository contains, for each result in this paper: the pre-registration file written before the run, the instrument that produced the number, the result file, and the fixed fold file every accuracy figure is scored against. The register in section 10 is checkable against it, which is the only reason that register is worth reading.

Status note. The two corrections that were in flight when this draft was first written have both landed, and both are reported in place rather than quietly absorbed: the honest-fold recomputation of section 7, which reinstated a withdrawn result, and the tie-corrected recomputation of every rank statistic in the programme (section 9), which moved 40 models by at most +0.009 and reordered none of them. The intervention result in section 3, the resolution floor in section 4 and the variance asymmetry in section 5 are comparisons on a fixed ruler and were unaffected by either.